

Correctness of the E-L* Noisy DFA Learner

(high-level, semi-formal)

1 Problem definition and preliminaries

1.1 The target and the noisy oracle

Definition 1 (Target). *we target a regular language $L \subseteq \Sigma^*$, recognised by a minimal DFA $A^* = (Q^*, \Sigma, t, q_0, F)$. Let $q^*(u)$ be the state that u is assigned by the DFA.*

One important property (Myhill–Nerode) is that two states are equal iff no suffix separates them:

$$q = q' \iff \forall w, \mathbf{1}[x_q w \in L] = \mathbf{1}[x_{q'} w \in L]$$

Definition 2 (Persistent Bernoulli oracle). *Let $s \in (0, \frac{1}{2}]$ be a signal level, with $\frac{1}{2}$ meaning a perfect signal. Let $f(x) \sim \text{Bernoulli}(s)$ be a random function where each $f(x)$ is independent. The oracle returns*

$$\text{MQ}(x) = \begin{cases} \mathbf{1}(1 \in L) & \text{if } f(x) = 1 \\ \mathbf{1}(1 \notin L) & \text{if } f(x) = 0 \end{cases}$$

Note that this means the noise is persistent: there is no way to re-query x gain more information.

Definition 3 (Access and goal). *The learner may only call MQ, but can call it on any string $x \in \Sigma^*$. It has access to a distribution D over Σ^* , and can sample from it. The goal is to output a DFA A that satisfies $\Pr_{x \sim D}[A(x) = A^*(x)] \geq 1 - \epsilon$ with probability $\geq 1 - \delta$.*

1.2 The midfix discrimination tree (noiseless)

Definition 4 (Midfix and cut). *A midfix is a string $m \in \Sigma^*$. Its cut is the set of states*

$$D_m = \{ q : x_q m \in L \},$$

which is well defined because $x_q m \in L$ is invariant to x by the Myhill–Nerode property.

Definition 5 (Midfix tree). *A midfix tree T is a binary tree whose internal nodes are labelled by midfixes and whose leaves are classes, that is, sets of states that the tree cannot distinguish.*

We route a state q down the tree by evaluating the cuts at each node m : if $q \in D_m$ we go left, else right. The leaf we reach is the class of q .

1.3 Accept-preserving suffixes

Definition 6 (Accept-preserving). *A suffix v is accept-preserving if it maps accept states to accept states and reject states to reject states, that is,*

$$\forall x, xv \in L \iff x \in L$$

Assumption 1 (Findable accept-preserving family). *Under the distribution that the learner can sample from, the probability that an accept-preserving suffix is drawn is at least $p_{\text{preserve}} > 0$.*

2 Denoising a cut with a suffix family

The learner cannot test $q \in D_m$ directly, the single read $\text{MQ}(x_q m)$ will have signal only s . Each node thus uses a family of suffixes $V = (v_1, \dots, v_k)$ and takes a vote over the k noisy queries $\text{MQ}(x_q m v_i)$.

Ideally the suffixes are accept-preserving (Def. 6), so that querying a string x against the family V gives k independent noisy reads of the same bit $\mathbf{1}[x \in L]$.

Definition 7 (Vote and decision). *The vote of a string x against the family is*

$$\bar{r}(x) = \frac{1}{k} \sum_{i=1}^k \text{MQ}(x v_i).$$

Given a split point β and an evidence margin τ , the decision is

$$\text{decide}_{\beta, \tau}(x) = \begin{cases} \text{ACCEPT} & \bar{r}(x) > \beta + \tau, \\ \text{REJECT} & \bar{r}(x) < \beta - \tau, \\ \text{INDECISIVE} & \text{otherwise.} \end{cases}$$

A node with midfix m classifies u by $\text{decide}_{\beta, \tau}(u m)$; the split β and margin τ are set by the clustering (§2.2), not fixed in advance.

If v are all accept-preserving, the k reads are independent with a common accept-probability p , so $\bar{r} \sim \frac{1}{k} \text{Bin}(k, p)$: a string with a definite side has $p = \beta \pm s$ for the per-read signal s , one with no definite side has $p = \beta$.

Definition 8 (fpr and fnr). *We can define*

$$\text{fpr} = \Pr_{p=\beta} [\text{decide}_{\beta, \tau} \in \{\text{ACCEPT}, \text{REJECT}\}], \quad \text{fnr} = \Pr_{p=\beta+s} [\text{decide}_{\beta, \tau} = \text{INDECISIVE}].$$

fpr upper-bounds the probability of a misclassification, while fnr is the probability of abstaining on a definite-side string.

The signal s is the one model parameter; the split β is estimated each round from the cut's own separation (Alg. 1). Given s and the current β , the pair (k, τ) is set to meet target budgets $\text{fpr} < \delta_{\text{fpr}}$ and $\text{fnr} < \delta_{\text{fnr}}$; the FNR budget is required of every prefix population separately (§2.1), not of their pooled average.

2.1 Prefix populations

The learner does not read its rates over one sample but over a collection of prefix populations, each drawn a different way and each held to the rates on its own:

- the uniform pool P_{unif} , sampled from the distribution D the goal is stated over;
- for each state q the current tree exposes, a population P_q of the prefixes the tree rests at q 's leaf;
- for each round, a boundary population of the strings that round's tree could not place decisively.

A prefix may lie in more than one. Keeping them apart is arithmetic: a rate over their union is carried by whichever population is largest, so a family that reads one small population entirely backwards moves the pooled rate by only that population’s share and hides there. Every rate below — the clustering loss, the FNR gate, the certification counts — is therefore stated per population. A population the learner cannot fill (a state no sampled string reaches) is dropped rather than held to a rate it has no prefixes to report.

2.2 The clustering algorithm

Reading each of n representative prefixes against each candidate suffix fills a noisy mask matrix $M[v][j] = \text{MQ}(x_j v) \in \{0, 1\}$, one row per candidate. The empty suffix’s row $M[\varepsilon][j] = \text{MQ}(x_j)$ is a noisy read of the true membership $\mathbf{1}[x_j \in L]$. A family is a triple (V, β, τ) : a size- k cluster V of rows anchored at ε , the split β estimated from its cut, and the evidence margin τ the calibration then sets. The clustering (Alg. 1) produces V and β ; the loop (Alg. 2) sets τ and grows the family until every population is decided and the cut is certified accept-preserving.

Algorithm 1 CLUSTER(β): the ε -anchored family and a refreshed split

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1:  $w_j \leftarrow 1/|P(j)|$  for each prefix  $j \triangleright P(j)$  its population: equal say per population, not per prefix
2:  $V \leftarrow \{\varepsilon\}, \ell^* \leftarrow \infty$ 
3: loop
4:    $c[j] \leftarrow \mathbf{1}[\text{mean}_{v \in V} M[v][j] > \beta]$   $\triangleright$  per-prefix centre
5:    $N \leftarrow$  the  $k$  rows of least loss  $\ell(v) = \sum_j w_j \mathbf{1}[M[v][j] \neq c[j]]$ 
6:   if  $\varepsilon \notin N$  or  $\sum_{v \in N} \ell(v) \geq \ell^*$  then break
7:    $V \leftarrow N, \ell^* \leftarrow \sum_{v \in N} \ell(v)$ 
8:    $\beta \leftarrow \frac{1}{2}(\bar{r}_{\text{acc}} + \bar{r}_{\text{rej}})$   $\triangleright$  midpoint of the mean vote on  $V$ ’s two sides
9: return  $(V, \beta)$ 
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The stopping test keeps ε from being evicted, not from being outvoted. A coherent group agreeing with ε on every state but one sits a hair from $M[\varepsilon]$, so ε stays a low-loss member while the other $k - 1$ rows cut that state the wrong way: the family is decisive, contains ε , and is not accept-preserving. The weighting is the first defence: a state whose prefixes are a percent of the union carries a percent of an unweighted loss, so a cut can invert it for free; weighted, that state has the same say as the pool and a cut against it costs. Two gates then read what the loss let through. The FNR gate holds each population to δ_{fnr} on its own and, when one reads high, grows that population — the lever that bears on the failure, since drawing more of a population it is not failing on only re-weights the average. But a cut in the wrong class is perfectly decisive, so the FNR gate passes it. The certification gate reads the cut back on $M[\varepsilon]$ — membership of x_j itself — and splits the judgement: any population may veto (it reads as the class it is not), only the uniform pool may admit (it separates the two classes at all), and admission takes positive evidence rather than absence of a veto.

2.3 Soundness of a decisive decision

Lemma 1 (A decisive decision matches the noiseless rule). *Any (V, β) induces a noiseless rule on states,*

$$\text{decide}^*(q) = \begin{cases} \text{ACCEPT} & \mu(q) > \beta, \\ \text{REJECT} & \mu(q) < \beta, \end{cases} \quad \mu(q) = \mathbb{E}[\bar{r}(x_q m)],$$

Algorithm 2 SAMPLEFAMILY: the resampling loop and its gates

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1: loop
2:    $(V, \beta) \leftarrow \text{CLUSTER}(\beta); \quad \tau \leftarrow \text{margin meeting the budgets at } (\beta, k)$ 
3:    $\widehat{\text{fnr}}_P \leftarrow \text{fraction of } P \text{ left INDECISIVE by } (V, \beta, \tau), \text{ each population } P$ 
4:    $P^* \leftarrow \arg \max_P \widehat{\text{fnr}}_P$ 
5:   if  $|V| < k$  or  $\widehat{\text{fnr}}_{P^*} > \delta_{\text{fnr}}$  then
6:     draw more suffixes, or grow  $P^*$ ; continue
7:   on  $M[\varepsilon]$ , count each population's two sides of the cut
8:   if some population reads as the opposite class then refuse and resample ▷ veto
9:   else if  $P_{\text{unif}}$ 's two sides both clear their class then
10:    return  $(V, \beta, \tau)$  ▷ admit
11:   else draw more of  $P_{\text{unif}}$  for the split ▷ undecided

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the true mean of the vote, a function of the state alone: a genuine Myhill–Nerode predicate. A decisive noisy decision matches it, $\text{decide}(um) = \text{decide}^*(q^*(u))$, except with probability $\leq \text{fpr}$.

Proof. Disagreeing — say ACCEPT while $\mu(q^*(u)) < \beta$ — needs $\bar{r}(um) > \beta + \tau$, a deviation of at least τ above the mean μ , of probability $\leq \text{fpr}$ whatever (V, β) . $\square$

A corollary is that soundness of ACCEPT/REJECT classifications (i.e., they reflect genuinely different classes) is guaranteed for any (V, β) . The choice of (V, β) only affects coverage: how many prefixes are classified decisively, and how many are left INDECISIVE. The gate holds the decided fraction $1 - \widehat{\text{fnr}}_P(V)$ to $\geq 1 - \delta_{\text{fnr}}$ on every population P , not merely on their pooled average, so no population is decided by proxy for another.

2.4 Convergence

Correctness does not rest on the clustering. As §2.2 notes, the ε -anchored cluster can settle — decisively, with ε retained — on a family that cuts one state against the language, and neither the anchoring nor the FNR gate sees it. What rules such a family out is the certification gate: the returned family is accept-preserving because it was certified so, not because the cluster found the right rows and not because a test failed to disprove it.

Lemma 2 (Certifying the cut, population by population). *Read the empty-suffix column $M[\varepsilon]$ over each population's two sides of the cut; where a side of population P holds n prefixes, h of them reading 1, one-sided binomial tests at level α apply. A population is read as the opposite class — a veto — when its accept side sits below $\beta + \tau$ or its reject side above $\beta - \tau$,*

$$\Pr[\text{Bin}(n_A, \beta + \tau) \leq h_A] \leq \alpha \quad \text{or} \quad \Pr[\text{Bin}(n_R, \beta - \tau) \geq h_R] \leq \alpha,$$

the level shared across the sides read. Any population may veto: a state whose prefixes all read backwards is disproof of accept-preservation whatever the pooled rate does. The uniform pool admits, and only it, when both its sides clear their own class,

$$\Pr[\text{Bin}(n_A, \beta + \tau) \geq h_A] \leq \alpha \quad \text{and} \quad \Pr[\text{Bin}(n_R, \beta - \tau) \leq h_R] \leq \alpha.$$

Only P_{unif} can, because admission is the claim that the family separates the two classes at all, and only a sample drawn at the oracle's base rate straddles both; a state's prefixes are one class and can veto but cannot attest to a separation they never straddle. The gate returns the family only when

P_{unif} admits and no population vetoes. Writing f for the share of an admitted side that is really the other class, its accept-side rate is $\beta + s - 2sf$, so admission certifies $f < (s - \tau)/2s$ on that side with confidence $\geq 1 - \alpha$. Absence of a veto is not admission: a family the uniform pool leaves undecided is resampled with more uniform reads, never returned.

Proof. Were an admitted accept side's true rate $\leq \beta + \tau$ — drift at or past the bar — then h_A is stochastically below $\text{Bin}(n_A, \beta + \tau)$, whose upper tail exceeds α except with probability $\leq \alpha$; likewise the reject side. So admission holds only for a cut whose two sides sit past $\beta \pm \tau$, which by rate $= \beta + s - 2sf$ is $f < (s - \tau)/2s$. The veto is the same test read the other way: a side sitting on the wrong side of its threshold past α is a population the oracle contradicts. A family the counts place on neither side of the bar triggers neither test and is resampled. $\square$

Theorem 1 (The gate returns a certified family). *Under Assumption 1 the loop halts with probability 1, and the family it returns decides at least $1 - \delta_{\text{fmr}}$ of every population and is admitted by the uniform pool with no population vetoing (Lemma 2): each side of its cut is a share $f < (s - \tau)/2s$ of the other class, with confidence $\geq 1 - \alpha$. When no accept-preserving family exists the uniform pool never admits and the idealized loop does not halt.*

Proof. The clustering proposes an ε -anchored family each round; the gate returns it only when every population is decided to δ_{fmr} , the uniform pool admits, and none vetoes, so the returned family carries those guarantees whatever the proposer did. It remains to see the loop halts under Assumption 1. Screened suffix draws are accept-preserving with probability $\geq p_{\text{preserve}}$; growing the worst population supplies the reads a state's rate needs rather than diluting it with a population that already passes; and as the pool and per-state populations grow the screen drops any suffix flipping a state on positive D -mass, so the candidate pool is eventually dominated by accept-preserving rows and the cluster proposes a driftless family. A driftless family decides each population and draws no veto, and the uniform pool admits it once its minority side is read out, so the loop returns. $\square$

When no accept-preserving family exists this loop runs forever; the implementation caps it with a give-up budget, spent only on families a population has vetoed (established drift, not mere uncertainty), and then reports the target has none.

The errors the per-population statement removes are both a rate standing in for a population it does not describe. A pooled FNR is carried by the largest population, so a family that leaves one state entirely indecisive — or cuts it entirely backwards — clears a pooled limit that the state's prefixes are a few percent of; stated per population the state reads 1 and the gate grows that state until the family resolves it. And admission read off a pooled column moves only by a drifted population's share: a family whose accept side was 69% mislabelled cleared a 10^{-3} threshold at 1.1×10^{-3} , because the aggregate barely moves under gross drift when the drifted class is thinly observed. Splitting the judgement fixes both — any population may veto, only the base-rate pool may admit, and each is held to the FNR on its own — and admission still takes positive evidence, so the gate fails closed. Soundness of the decisive labels (Lemma 1) is, as before, free of all of it.